

Assignment 1

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Instruction: (1) Do your work neatly. We will not evaluate if we don't understand what you have written. (2) Hard Deadline: 20/09/2026 Time: 11:59 pm. Anything submitted after the deadline will **NOT** be considered (3) Maximum Pts: 5+5

1. **Validity of the Parabolic Approximation:** Consider a conduction electron in silicon (lattice constant $a = 0.357$ nm) initially at rest near the conduction band minimum. A uniform electric field of $\mathcal{E} = 1.0 \times 10^4$ V/cm is applied along the [100] crystallographic direction. The electron undergoes lattice scattering with an average relaxation time of $\tau = 0.20$ ps.
 - (a) Calculate the wavevector change Δk gained by the electron between consecutive scattering events assuming the electrons always starts from rest after collision.
 - (b) Determine the wavevector magnitude at the Brillouin zone boundary along the [100] direction, defined by $k_{\text{BZ}} = \frac{2\pi}{a}$.
 - (c) Calculate the ratio $\frac{\Delta k}{k_{\text{BZ}}}$ as a percentage.
 - (d) Based on your result in part (3), explain why the parabolic energy band approximation ($E \propto k^2$) remains valid for low-field transport calculations in silicon.
2. **Hall Effect:** An experimental characterization technique used to determine the majority carrier type (n-type or p-type), carrier density, and mobility in bulk semiconductors. A rectangular semiconductor sample carries a current (I) along the (x)-direction. A magnetic field (B) is applied along the (z)-direction. Assume that the semiconductor has only one type of charge carrier (electron or hole) with concentration (c) and charge (q). Note that in case of electron the charge will be $-e$ and for hole it would be $+e$. Use the Lorentz force and basic relations for current density to answer the following for both if the carrier is electron or hole:
 - (a) Determine the direction of the magnetic force acting on a moving charge carrier. Explain why this causes charge accumulation on the sides of the sample and produces a transverse electric field, called the Hall field (E_H).
 - (b) At steady state, the transverse magnetic force is balanced by the electric force. Write this force-balance condition and hence obtain (E_H) in terms of the carrier drift velocity (v_d) and magnetic field (B).
 - (c) Using the relation between current density (J), carrier concentration (c), charge ($\pm e$), and drift velocity (v_d), eliminate (v_d) and obtain the Hall coefficient $R_H = \frac{E_H}{JB}$.
 - (d) If the width of the sample is (w), relate the Hall field to the measurable Hall voltage (V_H), and obtain an expression for (V_H) in terms of (I), (B), (n), (q), and (w).
 - (e) Explain very briefly how the sign of the Hall voltage can be used to distinguish an (n)-type semiconductor from a (p)-type semiconductor.